
ZK Compliance for cross-border Stellar Stablecoins.

# The Cross-Border Problem

### Issuer lock-in

Every anchor builds bespoke KYC rails — no portable credential layer across apps or corridors

### Credentials don't travel

Users re-verify per corridor, merchant, and payment flow — KYC never leaves the issuer silo

### PII follows payments

Customer documents re-sent with every cross-border transfer; travel rule checks stay manual

**Stablecoins need a portable issuer layer — verify once, prove per transfer, no document shipping.**

# The Solution

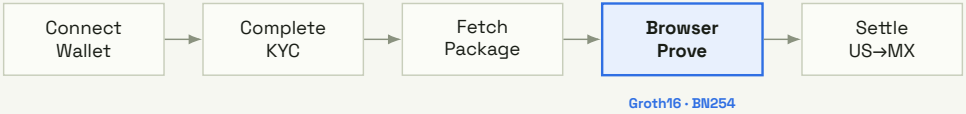
- **Verify once** — issuer KYC, Merkle root on-chain
- **Prove per transfer** — corridor rules in one Groth16 proof
- **Travel-rule binding** — source jurisdiction, destination registry, sanctions
- Soroban **verifies proof**, records **nullifier**, settles transfer

Checks per proof

**8**

One Groth16 snark

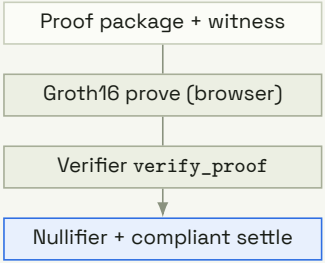
# How It Works




**One credential. Fresh proof for every cross-border transfer.**

# Groth16 on Stellar

- Custom Circom circuit — **8 compliance checks**
- **17 public inputs** rebound from on-chain state
- Dedicated **verifier contract** — BN254 pairing check
- `compliance.compliant_transfer` gates settlement
- **Nullifier** recorded — anti-replay per proof




Regulated compliance, not privacy — sender, recipient, asset, and amount stay public on-chain.

# Stellar ZK Primitives in Wisp

### BN254

Protocol 25 host functions. `bn254_multi_pairing_check` verifies Groth16 on Soroban.

### Poseidon2

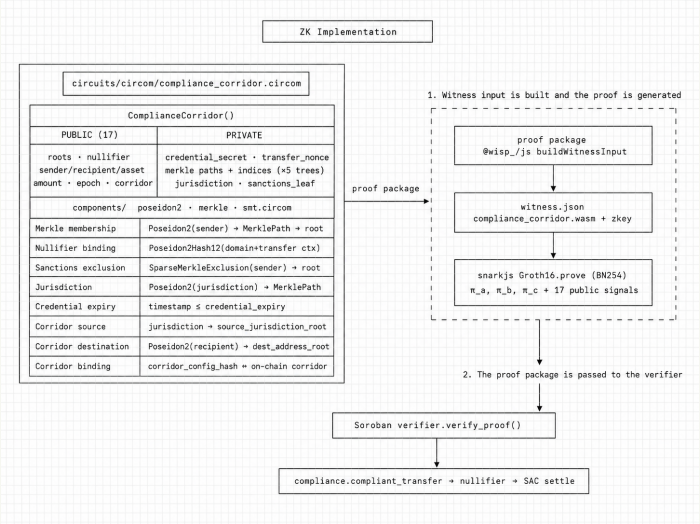
In-circuit hashing for Merkle trees, nullifiers, and corridor binding. Depth-20 trees.

### Travel rule

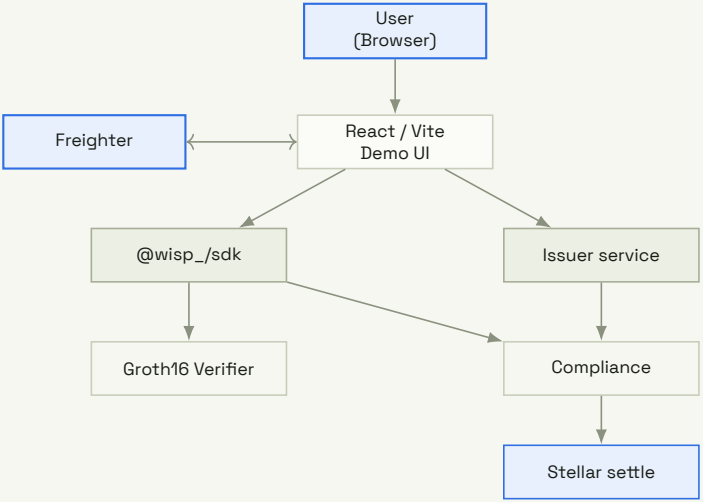
Corridor config hash binds source, destination, asset, and amount per route.

**Built for cross-border corridors · US→MX live on Stellar testnet**

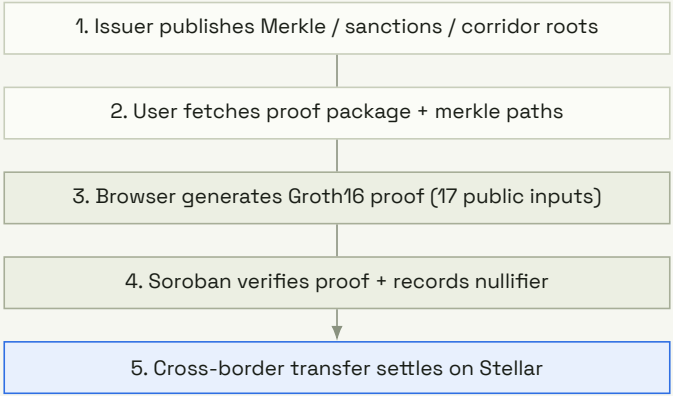
# ZK Architecture & Implementation



# System Architecture



# Transfer Lifecycle



# 8-in-1 Compliance Circuit

### Identity

KYC Merkle membership
Nullifier binding
Credential expiry

### Policy

Sanctions exclusion
Jurisdiction check
Corridor config hash

### Corridor

Source jurisdiction
Destination registry
US→MX live on testnet

**One proof enforces the full US→MX corridor — expand routes without new PII exposure**

# Live Demo Pipeline

**Live corridor: US→MX · wisp-zk.vercel.app/demo**

1. Connect Freighter wallet
2. Complete mock KYC onboarding
3. Fetch proof package from issuer
4. Generate Groth16 proof in browser
5. Settle cross-border transfer → audit `ProofSettled`

**Remittance corridor demo — tamper and replay rejection included**

# Built & Shipped

| 37+ | 8 | 5 | US→MX |
|---|---|---|---|
| Contract tests | Checks / proof | npm packages | Live corridor |

## Portable issuer stack

- `@wisp_/issuer-sdk` — register issuers, publish roots on-chain
- `@wisp_/sdk` — browser prove + Freighter wallet settle
- Travel-rule corridors with on-chain config hashes

## Live on testnet

- wisp-zk.vercel.app
- github.com/wurli-sh/wisp
- Demo video · First launch · Second launch

